



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

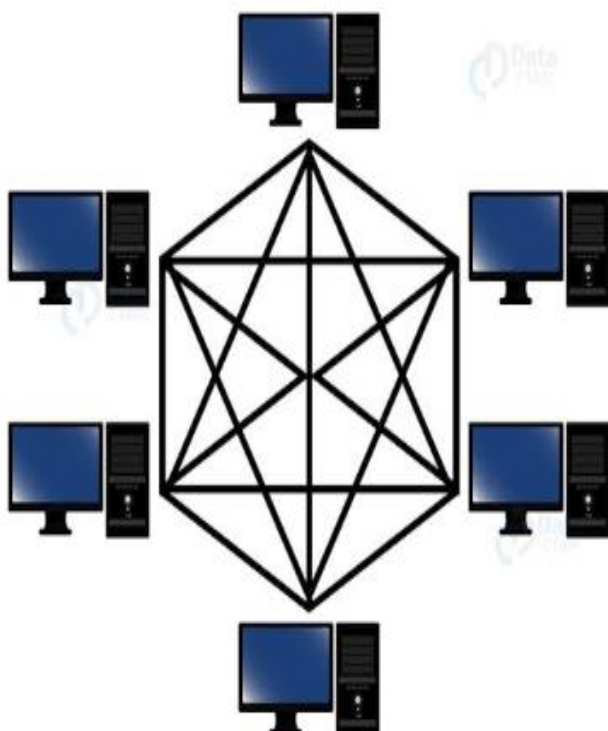
– Hira Kumar

Computer Networking



❖ Networking / Computer Network

- NET (जाल) + Working = Networking that means every things connect together. E.g. – Rail Network, Water/gas Pipeline network, Road network, in the same way computer network.



Q. What is Computer Network ?

- A computer network is a system that connects two or more computing devices to share data, information, resources, and services.
- These devices are linked using wired or wireless connections.
- Analog and digital signals are used for data communication in computer network but present (2026) all communication in digital forms.
- A computer network can be defined as a collection of nodes.
- A node can be any device capable of transmitting or receiving data.

➤ Advantages of Computer Network

1. **Resource Sharing** – A computer network allows users to share hardware resources, software, printers, storage devices, and Internet connections among multiple computers.
2. **Data and File Sharing** – It enables users to easily transfer, access, share, and update files and data between connected computers.
3. **Fast Communication** – A network provides quick and reliable communication through email, instant messaging, video conferencing, VoIP, and other communication services.
4. **Cost Reduction** – Sharing hardware, software, storage, and network resources reduces the overall installation, operational, and maintenance costs.
5. **Centralized Management** – Network administrators can centrally manage users, security, software, data, backups, and network resources, making administration easier.

➤ Disadvantages of Computer Network

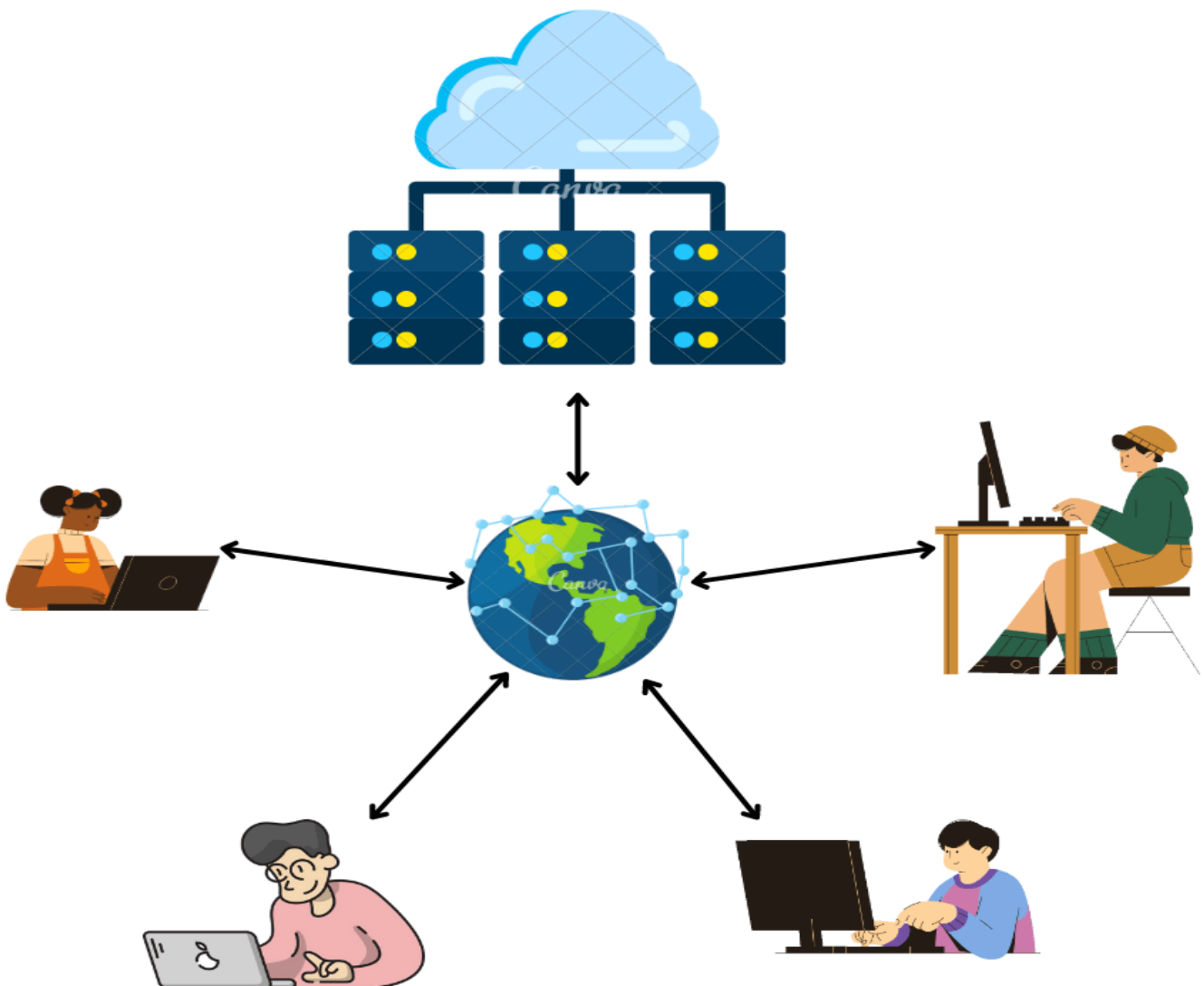
1. **Security Threats** – Computer networks are vulnerable to hacking, malware, viruses, unauthorized access, data theft, and cyberattacks.
2. **High Installation Cost** – Setting up a network may require cables, switches, routers, servers, network cards, software, and skilled technicians, resulting in high initial costs.
3. **Network Failure** – Failure of important network devices such as servers, switches, routers, or communication links can interrupt communication and access to shared resources.
4. **Maintenance and Administration** – Networks require regular

monitoring, troubleshooting, software updates, security management, and technical maintenance.

5. Privacy and Data Loss – Sharing data over a network can create risks of unauthorized access, privacy breaches, accidental deletion, and loss or modification of sensitive information.

Q. What is Internet ?

- Internet stands for interconnected network.
- Internet is a global network of interconnected computer networks that communicate and share information using standard protocols such as TCP/IP.
- Collection/ Interconnected of global network is called internet.
- The internet is a network of networks.
- Internet services is : WWW, Email, File transfer, Online gaming, VoIP, SSH/remote access.....etc.



➤ **Advantages of Internet**

- 1. Global Communication** – The Internet enables fast and easy communication with people around the world through email, instant messaging, voice calls, video calls, and social media.
- 2. Access to Information** – It provides access to a huge amount of educational, scientific, technical, historical, and general information from anywhere at any time.
- 3. Online Education** – Students can access online courses, digital libraries, video lectures, e-books, study materials, and virtual classrooms for learning and skill development.
- 4. Online Services and Transactions** – The Internet provides various services such as online banking, digital payments, online shopping, ticket booking, e-commerce, and government services.
- 5. Entertainment and Social Networking** – Users can enjoy online videos, music, movies, games, social media, and other digital entertainment services, as well as connect with friends and communities worldwide.

➤ **Disadvantages of Internet**

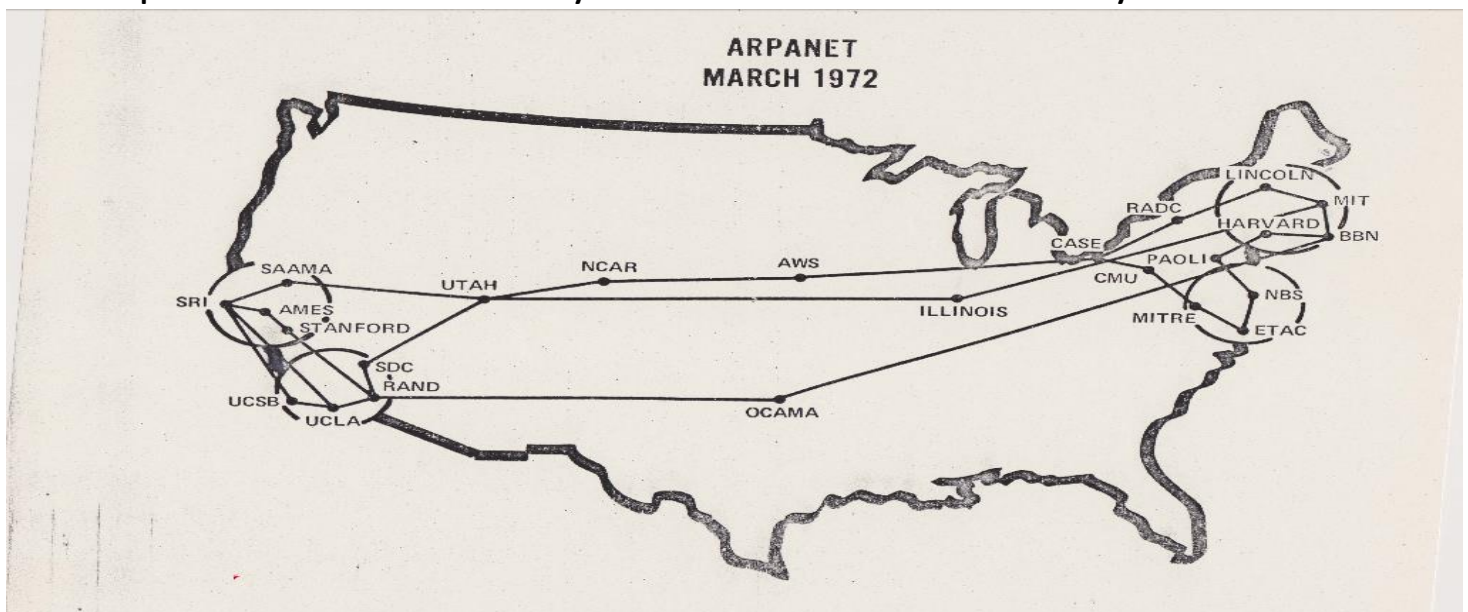
- 1. Cybersecurity Threats** – The Internet is vulnerable to hacking, malware, viruses, phishing, ransomware, and other cyberattacks that can damage systems or steal information.
- 2. Privacy and Data Theft** – Personal information such as passwords, financial details, photos, and browsing data may be collected, misused, or stolen by unauthorized persons.
- 3. Online Fraud and Scams** – Users may become victims of online banking fraud, phishing scams, fake websites, identity theft, and other Internet-based financial crimes.
- 4. Misinformation and Fake News** – The Internet allows false, misleading, or unverified information to spread rapidly, which can create confusion and mislead users.
- 5. Internet Addiction and Health Problems** – Excessive Internet use can lead to social media addiction, reduced productivity, sleep problems, lack of physical activity, and poor concentration.

❖ History of Network & Internet

- In 1958s, The U.S. government established ARPA (Advanced research projects agency). Its main purpose was to develop advanced technologies for defense and national security.
- In the 1960s, computers were becoming important for research, but computers at different universities and research centers could not easily communicate with each other, so not transfer any data.

After Research Agency name : ARP (Advanced research projects agency) of the US (United states) Department of Defense (DoD) wanted to develop a network that could connect computers at different locations and allow them to share information and resources.

- In the 1969, ARPA created ARPANET (Advanced research projects agency network).
- It was the world's first computer network and the precursor/after to the network.
- Initially , four major nodes were connected :
 - ✓ UCLA (University of California, Los angeles)
 - ✓ SRI (Stanford research institute – California, Menlo park)
 - ✓ UCSB (University of California, santa Barbara)
 - ✓ University of Utah
- ARPANET was a computer network developed by ARPA, a research agency of the U.S. department of defense, in 1969. Its first connection was b/w UCLA and SRI (distance 560-600 km). 1st message send both computer is "LOGIN" but system crash and receive only "LO".





- 1970s TCP/IP (Transmission control protocol / internet protocol) was developed by Vint Cerf and Bob Kahn.
- In 1972s ARPA was renamed DARPA (Defense advanced research projects agency).
- 1983 ARPANET adopted TCP/IP and the internet was officially born.
- 1989 Tim Berners-Lee developed the WWW (World wide web), a system of hyperlinked documents that could be accessed through internet.

✓ **Key Points**

The first public internet services in India was introduced by VSNL (Videsh Sanchar Nigam Limited) in 1995 and 2002 tata group acquisition and its called Tata Communications.

❖ **Components of Computer Network**

Main components is 6 -

- 1. Sender (Source)** – The device that creates and sends data over the network, such as a computer or smartphone.
- 2. Receiver (Destination)** – The device that receives the transmitted data, such as a computer, server, or printer.

3. Transmission Medium – The communication path through which data travels, such as twisted-pair cable, coaxial cable, fiber-optic cable, or wireless signals.

4. Network Interface Card (NIC) – A hardware component that connects a computer or other device to a network and enables network communication.

5. Protocols – A set of rules and standards that control how data is formatted, transmitted, received, and interpreted. Examples include TCP/IP, HTTP, FTP, and Ethernet.

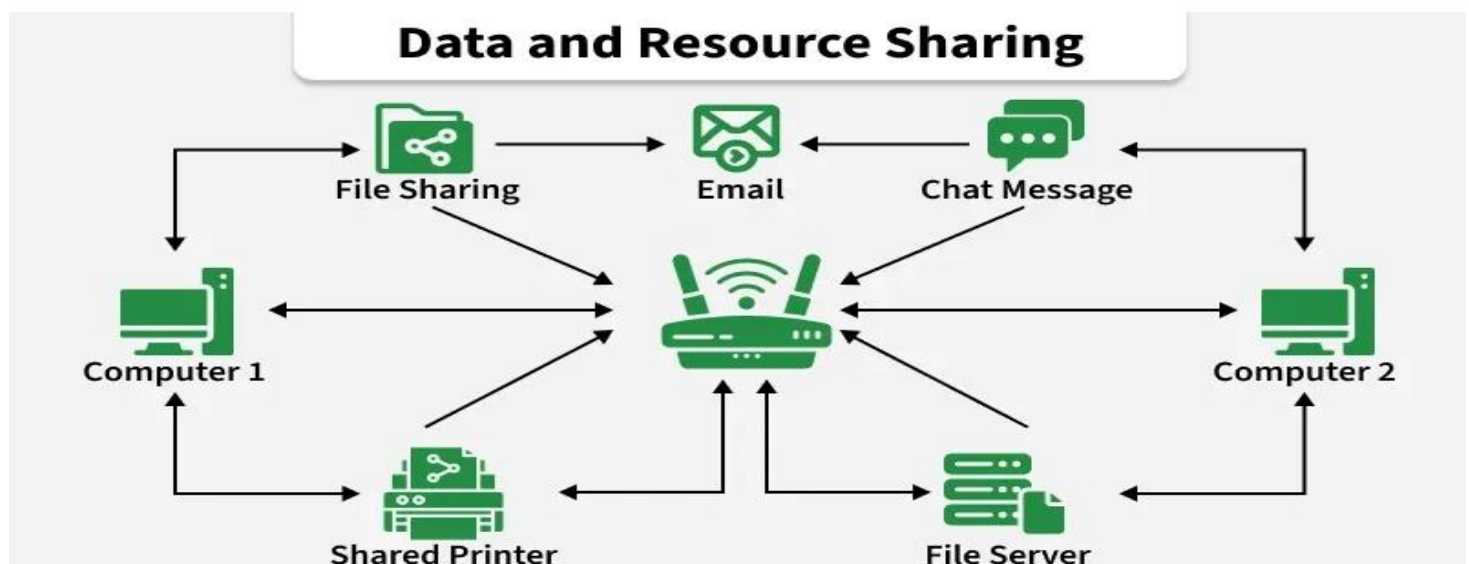
6. Network Software – Software such as network operating systems, drivers, and network management tools that helps operate, control, and manage the network.

7. Network Devices – Devices such as switches, routers, hubs, bridges, repeaters, and gateways that connect devices and manage/direct data transmission. (not mandatory for a two node network)

❖ Application of computer networks:

1. Resource Sharing

Resource sharing is an application of a computer network. Resource sharing means you can share one Hardware and Software among multiple users. Hardware includes printers, Disks, Fax Machines, etc. Computing devices. And Software includes Atom, Oracle VM Virtual Box, Postman, Android Studio, etc.



2.Information Sharing

Using a Computer network, we can share Information over the network, and it provides Search capabilities such as WWW. Over the network, single information can be shared among the many users over the internet.

3.Communication

Communication includes email, calls, message broadcast, electronic funds transfer system etc.

4.Entertainment Industry

In Entertainment industry also uses computer networks widely. Some of the Entertainment industries are Video on demand, Multiperson real-time simulation games, movie/TV programs, etc

5.Access to Remote Databases

Computer networks allow us to access the Remote Database of the various applications by the endusers. Some applications are Reservation for Hotels, Airplane Booking, Home Banking, Automated Newspaper, etc.

6.Home applications

There are many common uses of the computer network are as home applications.e.g – Smart TV, CCTV, Light,AC,CAR,Washing machine,.....all IOT device.

7.Business applications

The result of business application here is resource sharing. And the purpose of resource sharing is that without moving to the physical location of the resource, all the data, plans and tools can be shared to any network user.

8.Social media

Social media is also a great example of a computer network application. It helps people to share and receive any information related to political, ethical and social issues.

❖ Types of Network

(Based on Geographical area)

Many types of network -

1. LAN - Local Area Network
2. WLAN - Wireless Local Area Network
3. WAN - Wide Area Network
4. MAN - Metropolitan Area Network
5. SAN - Storage Area Network, System Area Network, Server Area Network, or sometimes Small Area Network
6. CAN - Campus Area Network, Controller Area Network, or sometimes Cluster Area Network,
7. PAN - Personal Area Network
8. DAN - Desk Area Network
9. VPN – Virtual Private Network

Exam point of view only 4 network is important

$PAN < LAN < MAN < WAN$

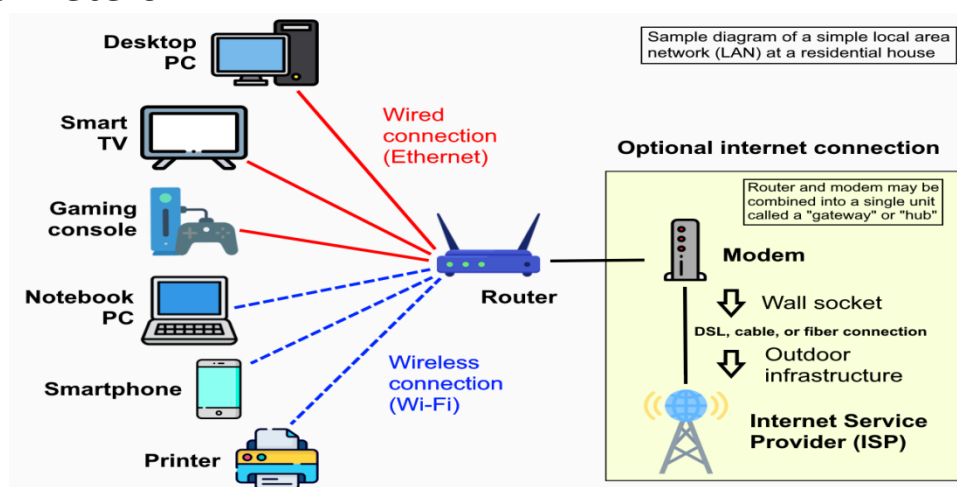
1. PAN: Personal area networks appropriate to personal or separate workspace under the range of 10 meters. These networks are mostly used to connect tablets, smartphones and laptops.

- ✓ **Network:** Private network
- ✓ **Area:** Very small area around a person
- ✓ **Use:** Connect personal devices
- ✓ **Transfer Speed:** Up to several Gbps
- ✓ **Devices:** Smartphone / Laptop / Smartwatch
- ✓ **Security:** High security
- ✓ **Cable:** USB cable / No cable (wireless)
- ✓ **Types:** Wired & Wireless
- ✓ **Range:** 1 – 10 meters



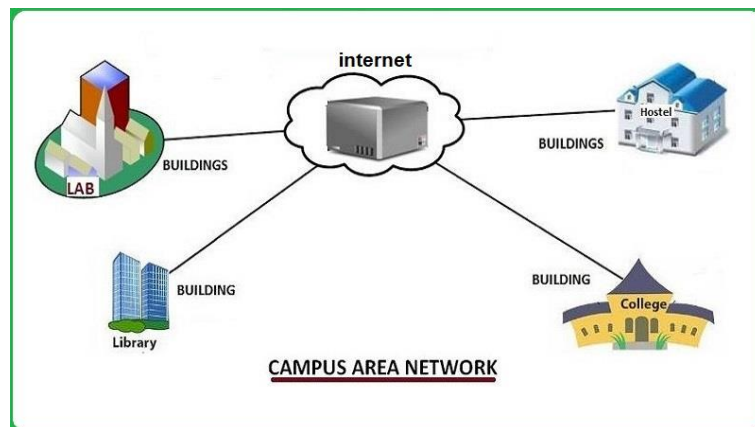
2. LAN: Local area networks are mainly used to connect personal devices within a few kilometers of a limited area. These networks are used in offices, companies and factories to exchange data and Information.

- ✓ **Network:** Private network
- ✓ **Area:** Small geographical area
- ✓ **Use:** Share resources
- ✓ **Transfer Speed:** 10 Mbps to 1 Gbps
- ✓ **Devices:** Hub / Switch
- ✓ **Security:** High security
- ✓ **Cable:** Twisted-pair cable / Coaxial cable
- ✓ **Types:** Wired & Wireless
- ✓ **Range:** 100 meters – 2 km



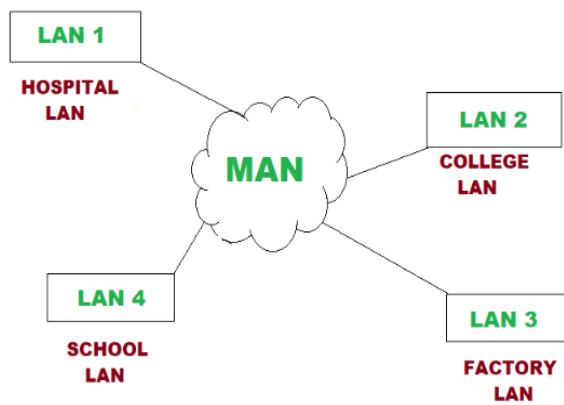
3. CAN: Campus area networks are used to connect limited geographic areas. CAN interconnect multiple local area networks (LAN) within Colleges, Universities, Corporates buildings, etc.

- ✓ **Network:** Private network
- ✓ **Area:** Campus or organization
- ✓ **Use:** Connect multiple LANs within a campus
- ✓ **Transfer Speed:** 100 Mbps to 10 Gbps
- ✓ **Devices:** Switches / Routers
- ✓ **Security:** High security
- ✓ **Cable:** Fiber-optic / Twisted-pair cable
- ✓ **Types:** Wired & Wireless
- ✓ **Range:** 1 – 5 km



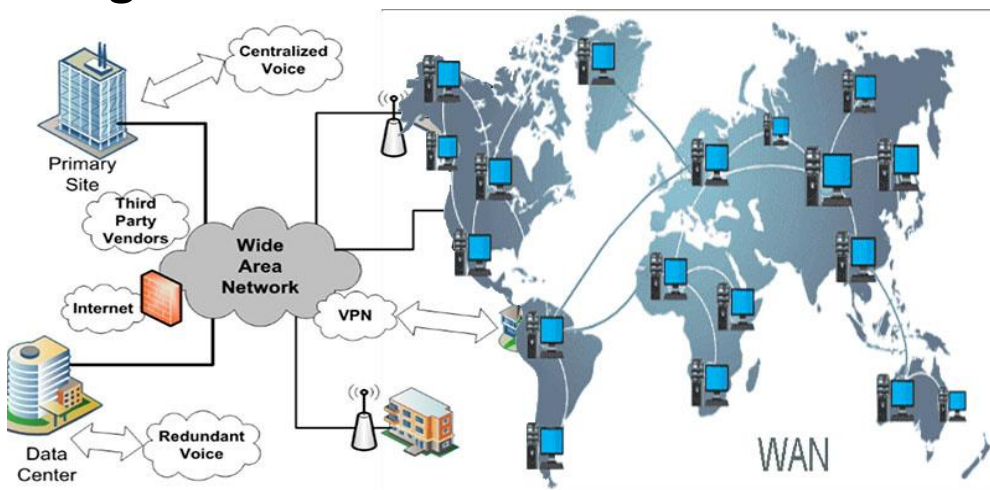
4. MAN: Metropolitan area networks are used to connect the devices over an entire city under the range of up to 50 km. These networks are used in the telephone company network and cable TV network.

- ✓ **Network:** Private or Public network
- ✓ **Area:** City or metropolitan area
- ✓ **Use:** Connect multiple LANs within a city
- ✓ **Transfer Speed:** 10 Mbps to 10 Gbps
- ✓ **Devices:** Routers / Switches
- ✓ **Security:** Medium to high
- ✓ **Cable:** Fiber-optic / Coaxial cable
- ✓ **Types:** Wired & Wireless
- ✓ **Range:** 5 – 50 km



5. WAN: Wide Area Networks are used in the wide geographical range over a country and continent. These networks are used in military services, mobile operators, railways and airlines reservations.

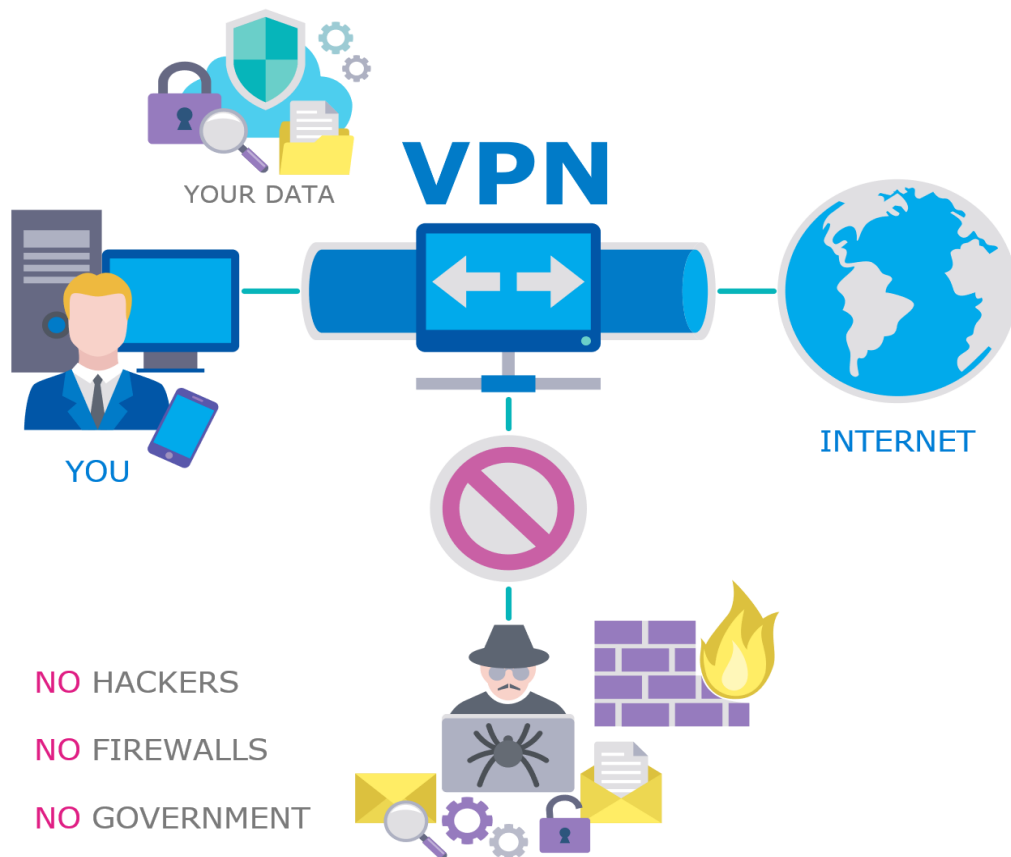
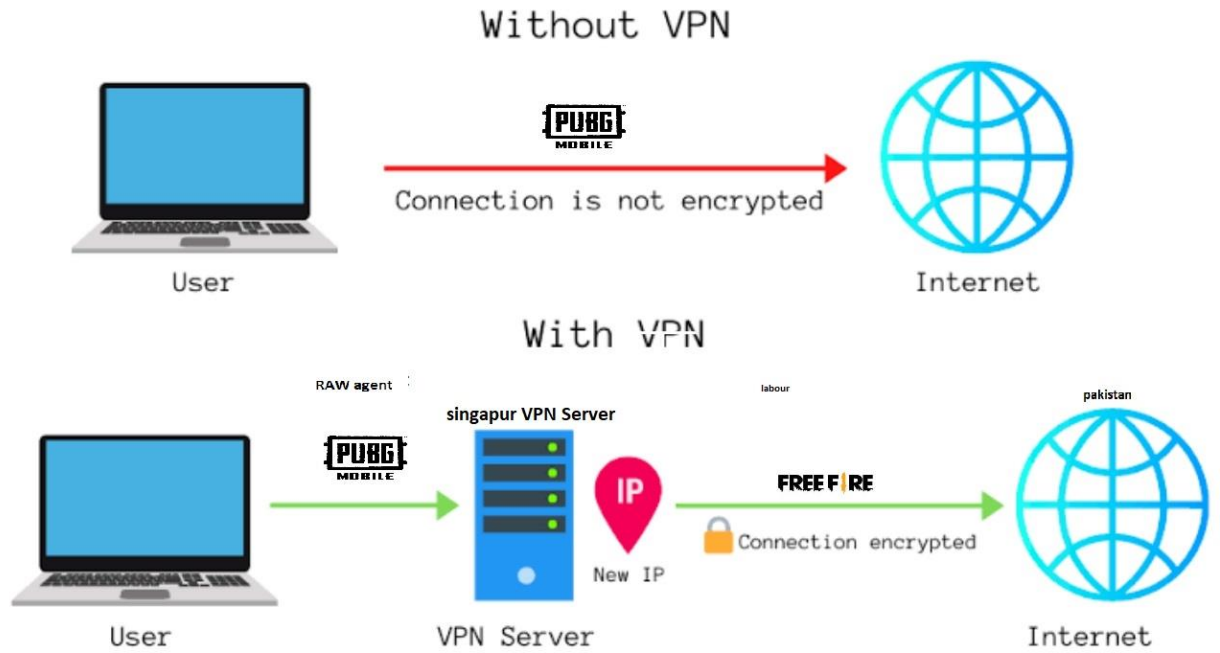
- ✓ **Network:** Private or Public network
- ✓ **Area:** Large geographical area
- ✓ **Use:** Connect LANs/MANs across cities, countries or continents
- ✓ **Transfer Speed:** Mbps to Gbps
- ✓ **Devices:** Routers / Modems / Switches
- ✓ **Security:** Relatively lower than LAN
- ✓ **Cable:** Fiber-optic / Undersea cable / Wireless links
- ✓ **Types:** Wired & Wireless
- ✓ **Range:** 50 km to worldwide



6. VPN :- A Virtual Private Network is a type of network that is created by using the Internet, to connect to another private network, such as a company's internal network or other country VPN server.

This is a fast spreading network nowadays, whose use has increased rapidly in big organizations. These networks are virtual as well as

private, private because in this network the privacy of an organization is completely guaranteed and virtual because this network does not use WAN.



❖ PAN, LAN, CAN, MAN and WAN – Network Comparison

Point	PAN	LAN	CAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Network	Private network	Private network	Private network	Private or Public network	Private or Public network
Coverage Area	Around a person	Building / small area	Campus / organization	City	Country / continent / worldwide
Main Purpose	Connect personal devices	Share resources	Connect LANs in a campus	Connect LANs across a city	Connect networks over large distances
Transfer Speed	Up to several Gbps	10 Mbps to 1 Gbps	100 Mbps to 10 Gbps	10 Mbps to 10 Gbps	Mbps to Gbps
Devices	Smartphone / Laptop / Smartwatch	Hub / Switch	Switches / Routers	Routers / Switches	Routers / Modems / Switches
Security	High	High	High	Medium to High	Relatively lower than LAN
Cable / Medium	USB / Wireless	Twisted-pair / Coaxial / Fiber / Wi-Fi	Fiber / Twisted-pair / Wi-Fi	Fiber / Microwave	Fiber / Undersea cable / Satellite / Microwave
Types	Wired & Wireless	Wired & Wireless	Wired & Wireless	Wired & Wireless	Wired & Wireless
Range	1–10 meters	100 meters – 2 km*	Campus-sized area	5–50 km*	50 km to worldwide*
Ownership	Personal	Private	Private	Private/Public	Private/Public
Common Technologies	Bluetooth, USB, Zigbee	Ethernet, Wi-Fi	Ethernet, Wi-Fi, Fiber	Fiber, Ethernet, Microwave	MPLS, leased lines, Internet
Cost	Very low	Low	Medium	High	Very high
Maintenance	Easy	Easy	Moderate	Difficult	Complex
Example	Phone + Smartwatch	Computer lab	University campus network	City-wide network	Internet

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